

AnyCam 3.3.0-rc1.0 Audit Report

Release audit - 04 October 2026 - one camera connection through go2rtc - tag 3.3.0-rc1.0

Scope: phase 3.3.0 of the build plan: C4 and B15. CrystalHeeler approved C4 on 2026-10-04 with this security condition: go2rtc's RTSP server on, listening on 127.0.0.1 only, with a random password set at each start. Built in the untested run on top of 3.2.0-rc1.0. This build is not field-tested.

1. Security review of the RTSP server

Control	Before 3.3.0	3.3.0
RTSP server	Off (listen "")	127.0.0.1:28554 only
Login	-	User anycam, a 24-byte random password from secrets, new at each add-on start; never written to disk or sent to the page
Reach	-	Nothing outside the Pi: loopback only. On the Pi, a program needs the password, which exists only in the add-on's memory and in its ffmpeg arguments
Modules	api, ws, rtsp, webrtc, mp4	Unchanged: no exec, echo, expr or ffmpeg
API	127.0.0.1 only	Unchanged
Release gate	Pins listen ""	Pins the 127.0.0.1 listen address and the user and password keys

Remaining exposure: a process on the Pi that can read the add-on's process arguments can read the password. Such a process can already read the camera passwords in the same arguments, as before 3.3.0. Log lines pass the password filter, which removes the user and password from every URL.

2. Who reads through go2rtc

User	Before	3.3.0
Live card, Enhanced View live	go2rtc	go2rtc, same stream names as the users below for the same source
Card pictures and classic view (snapshot loop)	Camera	go2rtc's copy, always over TCP
Motion detector (small stream)	Camera	go2rtc's copy
Recording buffer, and a recording without it (main stream)	Camera	go2rtc's copy
HTTP snapshots, the H.265+ fallback, probes	Camera	Camera (not RTSP streams held open)

- One go2rtc stream per source: `_go2rtc_shared_name` hashes the camera id and the address without its password. A name never changes its source, so no user's stream is swapped under it, and the name that reaches the browser carries no password information.
- go2rtc dials the camera when the first user connects and holds one connection for all of them.
- Way back: a run with no picture or data counts against the camera; 3 in a row and AnyCam opens the camera directly again until the add-on restarts. A good run clears the count. go2rtc not running, or a stream it does not

accept: direct at once.

3. B15

B15 was held back in 2.6.5 because keeping the card's stream while the classic view starts needed two camera connections, which rate-limited cameras and DVRs refuse. With C4 the classic view's ffmpeg reads go2rtc's copy: no second camera connection, and the camera stream stays open in go2rtc while the live card or motion detection use it. The snapshot loop itself still restarts at full resolution, as before.

4. Recording

The build plan named go2rtc's MP4 output for recordings. The recording buffer ffmpeg stays instead: it holds the 3 s before the motion, which go2rtc's MP4 output does not. It now reads go2rtc's copy, so the goal of C4, one camera connection, holds.

Tests run

Suite	Result	Scope
Release gate	PASS	Eight gates; 15 files; build inputs checked online; the go2rtc contract updated.
Python integration	PASS	513 of 513; 10 new in section AD against the fake go2rtc; the config check now expects the RTSP server.
JavaScript behaviour	PASS	139 of 139, unchanged.
Undefined names	PASS	15 files, 0.
Not run	NOTED	No field test (CrystalHeeler's order). The image was not built. A real go2rtc serving RTSP to ffmpeg did not run: the tests use a fake go2rtc and a fake ffmpeg.

Best-practices compliance

Measured against *docs/AnyCam_Coding_Best_Practices.md*, Parts 1 to 6. Rows this change touches; the rest carry from the 3.2.0-rc1.0 audit.

Standard	Check applied	Result	Note
1.1 Truncated function	AST, contracts, undefined names	PASS	15 files; the undefined-name check caught one slip.
1.10 Exceptions	Failure paths	PASS	A refused stream or go2rtc down means a direct connection.
1.18 Testing	New tests	PASS	10 server checks.
6.1 Security	New listener	PASS	Loopback only, random password per start, pinned by the gate.
Rule 9 (privacy)	New text	PASS	Tests use made-up addresses only.

Known issues carried forward

- Recordings keep their buffer ffmpeg, now fed by go2rtc.
- **B6:** the Microseven waits until CrystalHeeler unlocks it.
- **C21:** the classic engine stays as the fallback.

Acceptance (field test)

- The log shows no ffmpeg reading a camera address directly while go2rtc runs (only 127.0.0.1:28554).
- Test system B: an armed DVR channel records, with the 3 s before the motion; the classic view opens while the card plays.
- Test system A: the Hikvision PTZ card, Enhanced View and motion together.
- From another computer, port 28554 on the Pi does not answer.

Files changed

File	Change
anycam_go2rtc.py	RTSP server config; _go2rtc_shared_name, _go2rtc_relay, _go2rtc_relay_result; the card uses the shared name.
anycam_focus.py	Enhanced View uses the shared name.
anycam_snap.py, anycam_motion.py	ffmpeg reads through _go2rtc_relay.
camera_discovery.py	Security model note; version 3.3.0-rc1.0.
verify_release.py	The go2rtc contract.
tests/	Section AD; the config check; go2rtc off after its tests.
CHANGELOG.md, config.yaml, docs	3.3.0-rc1.0; build plan.